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[22TBSCSMA52]

FIFTH SEMESTER B.Tech EXAMINATION, NOVEMBER-2024

STATISTICS AND EXPLORATORY DATA ANALYTICS

Time: 3 Hours **Maximum Marks: 100**

Instructions:
i. **Missing data may be suitably assumed**
ii. **Answer all Questions**

		PART-A	**5 x2=10**		
1.	**a.**	Calculate the trimmed mean at 10% for the data set: 5, 4, 6, 8, 10, 11, 0, 7, 18.	L2	CO1	**02**
	b.	Illustrate how Chi- square test is different from F test.	L2	CO1	**02**
	c.	Demonstrate how to calculate RMSE using the given data. Actual values y=[3,−0.5,2,7], Predicted values =[2.5,0.0,2,8]	L2	CO4	**02**
	d.	Explain any two differences between a List and a Tuple.	L2	CO4	**02**
	e.	Explain use of pd.crosstab function in Python.	L2	CO2	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Interpret the variance and standard deviation of the following observations about the mean: 17.05, 17.02, 17.0,17.25,17.04, 17.34, 17.07, 17.21	L3	CO1	**05**
		OR			
	b.	Following table shows the age distribution of persons in a particular region:	L3	CO1	**05**

Age	Below 10	Below 20	Below 30	Below 40	Below 50	Below 60	Below 70	70 & above
No of persons	2	5	9	12	14	5	15.5	15.6

Calculate median. Interpret why Median is a more suitable measure of central tendency in this case?

3.	**a.**	Given a binary classification problem with a confusion matrix defined as: $\begin{bmatrix} 80 & 90 \\ 10 & 90 \end{bmatrix}$ Calculate the F1-Score for this classifier.	L3	CO3	**05**
		OR			
	b.	Demonstrate how Correlation is different from Causation using suitable example.	L3	CO3	**05**
4.	**a.**	Find the linear regression equation for the given data:	L3	CO1	**05**

X	7	9	3	8
Y	8	4	4	2

OR

	b.	Interpret the following situation. In one sample of 8 observations from a normal population, the sum of the squares of deviations of the sample values from the sample mean is 84.4 and in another sample of 10 observations it was 102.6. Test at 5% level whether the populations have the same variance. F (7,9) at 0.05 level = 3.29.	L3	CO1	**05**
5.	**a.**	Two random samples were drawn from two normal populations and their values are:	L3	CO6	**05**

S1	20	16	26	27	23	22	18	24	25	19		
S2	27	33	42	35	32	34	38	28	41	43	30	37

Analyze whether the two populations have the same variance at the 5% level of significance. Given: $F_{0.05}(9,11)=2.9$.

		OR			
	b.	Explain the difference between Multiple regression and Polynomial regression.	L3	CO6	**05**
6.	**a.**	Illustrate the different data structures available in Pandas with suitable examples.	L3	CO5	**05**
		OR			
	b.	Illustrate the key features of Plotly Express using suitable examples.	L3	CO5	**05**
7.	**a.**	Demonstrate any two DATETIME LIBRARY with suitable code segments.	L3	CO4	**05**
		OR			
	b.	Demonstrate pd.crosstab function in Python.	L3	CO4	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	Students of a class were given an aptitude test. Their marks were found to be normally distributed with mean 60 and standard deviation is 5. What percentage of students scored (i) more than 60 marks, (ii) less than 56 marks? Given $P(0<Z<0.8)=0.2881$.	L3	CO1	**10**
		OR			
	b.	The height of adult males in a certain population is normally distributed with a mean of 175 cm and a standard deviation of 10 cm. What is the probability that a randomly selected male from this population has a height between 160 cm and 185 cm? Given $P(Z\leq 1.0) = 0.8413$, and $P(Z \leq -1.5) = 0.0668$	L3	CO1	**10**

9.	**a.**	Explain Type-I and Type-II errors. Employ a suitable test to accept the nutritionist claim that the average daily calorie intake for adults is 2500 calories. A random sample of 15 adults shows an average daily intake of 2400 calories with a standard deviation of 100 calories. Is there enough evidence at the 5% significance level to reject the nutritionist's claim? (critical value ± 2.145)	L3	CO3	**10**
		OR			
	b.	Explain One-Tailed and Two-Tailed Test. A factory claims that the mean weight of a pack of sugar it produces is 1 kg. A random sample of 25 packs is taken, and the sample mean is 0.98 kg with a standard deviation of 0.05 kg. Test the claim at a 5% significance level. The t-value at 0.05 level for 24 degrees of freedom is ± 2.064.	L3	CO3	**10**
10.	**a.**	Three different kinds of food are tested on three groups of rats for 5 weeks. The objective is to check the difference in mean weight (in grams) of the rats per week. Apply one-way ANOVA using a 0.05 significance level to the following data:	L3	CO6	**10**

Food1	8	12	19	8	6	11
Food2	4	5	4	6	9	7
Food3	11	8	7	13	7	9

F tabulated =2.76

		OR			
	b.	In an antimalarial campaign, quinine was administered to 500 patients out of a total population of 2000. The number of fever cases as follows:	L3	CO6	**10**

Treatment	Fever (A)	No fever (a)	Total
Quinine (B)	140 (AB)	30 (aB)	170 (B)
No Quinine (b)	60 (Ab)	20 (ab)	80 (b)
Total	200(A)	50 (a)	250 (N)

Discuss the usefulness of quinine in checking malaria.

Given: For df =1, $\chi^2_{0.05} = 3.84$

11.	**a.**	If a sample of 400 male workers has a mean height of 67.47 inches, is it reasonable to regard the sample as a sample from a large population with a mean height of 67.39 inches and a standard deviation of 1.30 inches at a 5% level of significance? p-value = 0.109	L3	CO4	**10**
		OR			
	b.	A researcher wants to test if there is a significant difference in the average hours studied per week between two groups of students from two different universities. A random sample from both universities yields the following data:	L3	CO4	**10**

		Sample 1: Sample size (n_1) = 50 Sample mean (x_1) = 35hrs Population sd (σ_1) = 8 **Sample 2:** Sample size (n_2) = 60 Sample mean (x_2) =30hrs Population sd (σ_1) = 10 Conduct Using two-tailed test, examine whether the means are different. For a two-tailed test at a 5% significance level, the critical z-values are ±1.96			
12.	**a.**	Explain Data Frame in Python? Write the key features of it, and write code segments to create, to display data type, to select columns, to filter data in a data frame.	L3	CO2	**10**
		OR			
	b.	Demonstrate how manipulation of textual data in Python can be done using various built-in methods and libraries.	L3	CO2	**10**
13.	**a.**	Explain Pandas Data Selection, Filtering, Manipulation, Cleaning, and Grouping LIBRARY FUNCTIONS in detail with code segments.	L3	CO5	**10**
		OR			
	b.	Illustrate Statistical Analysis with Seaborn Library FUNCTIONS in detail with code segments.	L3	CO5	**10**